

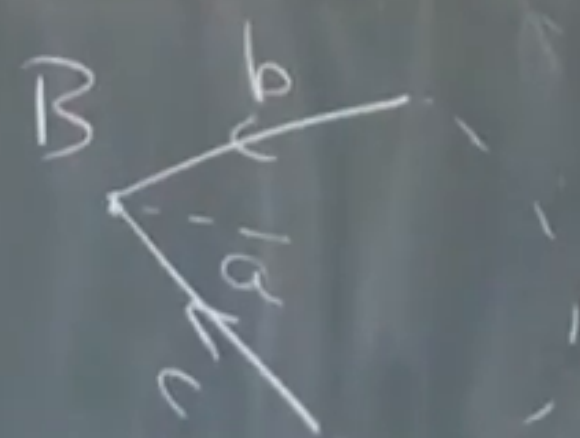
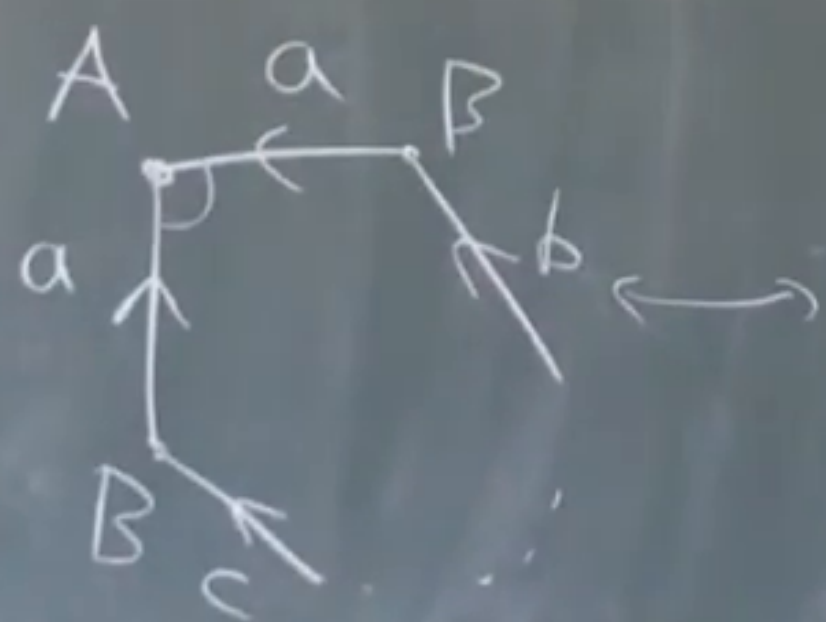
Euler number of a comb. surface:

Thm $\chi = V - E + F$ is well defined on
a combinatorial surface.

Note: $V = \# \text{ vertices / types / equivalence classes}$.

Pl

①



$$\boxed{\chi \rightarrow \chi}$$

$$V \rightarrow V - 1$$

$$E \rightarrow E - 1$$

$$F \rightarrow F$$